

**Earth: The Water Planet — Where the Water Is** S6E3a

About **71% of Earth's surface** is covered by water — the "blue marble." But most of it is NOT drinkable:

SALT WATER (oceans) — 97%

FRESH — 3%

Type / Location	About How Much	Notes
Oceans (salt water)	97% of all water	Too salty to drink or water crops
Ice caps & glaciers	About 2%	Frozen fresh water locked at the poles & mountains — most of the fresh water!
Groundwater	Most of the rest	Fresh water stored underground in soil & rock (aquifers — towns pump it up with wells)
Rivers, lakes & swamps	A tiny fraction	The surface fresh water we most easily use

😲 **Surprising but true:** less than **1%** of all Earth's water is fresh water people can easily reach — that's why protecting rivers, lakes, and groundwater matters so much.

**ACTIVITY 1: Where's the Water?**

- _____ % of Earth's surface is covered by water
- _____ % of Earth's water is salty ocean
- _____ % of Earth's water is fresh
- Most FRESH water is stored as: _____
- Underground water layer: _____
- Easiest fresh water to use: rivers & _____

**The Water Cycle** S6E3b

Water moves in a never-ending loop between oceans, air, and land. The energy source powering it all: **THE SUN**.

Evaporation → Condensation → Precipitation → Collection / Runoff → repeat!

Step	What Happens	Driven By
Evaporation	The Sun heats water → it becomes invisible water vapor (a gas) that rises	Sun's heat
Condensation	High up where it's cooler, vapor turns back into tiny droplets — clouds form	Cooling
Precipitation	Droplets join, grow heavy, and fall as rain, snow, sleet, or hail	Gravity
Collection / Runoff	Water gathers in oceans, lakes & rivers or soaks into the ground — then repeats	Gravity

★ **The Sun is the engine:** the Sun lifts water into the sky; **gravity** brings it back down. 🌿 **Transpiration** = plants releasing water vapor from their leaves (also adds water to the air).

**ACTIVITY 2: Name That Step!**

- Clouds forming from cooling vapor: _____
- Liquid water turning into rising vapor: _____
- Rain, snow, sleet, or hail falling: _____
- Water gathering in lakes and oceans: _____
- Plants releasing vapor from leaves: _____
- The energy source that powers it all: _____

💧 Flip over for the ocean floor, waves & currents, and tides! →

 **The Ocean Floor** S6E3c

The ocean bottom is a hidden landscape with mountains, valleys, and plains. Scientists map it with **sonar** (sound waves) and satellites.

Feature	What It Is
Continental Shelf	The shallow , gently sloping edge of a continent, just offshore
Continental Slope	The steeper drop-off where the shelf ends and the deep ocean begins
Abyssal Plain	The wide, flat floor of the deep ocean
Mid-Ocean Ridge	A chain of underwater mountains running along the sea floor
Trench	A deep, narrow valley — the deepest parts of the ocean

Deeper than Everest is tall: the Mariana Trench is ~11 km deep — drop Mount Everest in and its peak would still be over a kilometer underwater!

ACTIVITY 3: Ocean Floor Tour

Match each clue to a feature (shelf, slope, abyssal plain, ridge, trench):

- Underwater mountain chain: _____
- The deepest part of the ocean: _____
- Shallow edge right next to a continent: _____
- The steep drop-off into deep water: _____
- Wide flat floor of the deep ocean: _____
- Tool that maps the floor with sound: _____

Waves & Currents — Not the Same Thing! S6E3d

WAVE		CURRENT
What moves	Energy (the water mostly stays put, bobbing up & down)	The water itself , flowing in one direction — a "river in the ocean"
Main cause	Wind blowing on the surface	Wind (surface) + temperature & saltness (deep: cold salty water sinks, warm rises)
Effect	Shapes shorelines, makes surf	Moves heat around the planet → affects climate

Stadium wave: in a crowd wave, people stand and sit but don't run around the stadium — energy moves, people stay. Ocean waves work the same way!

Tides — the Moon Pulls the Ocean S6E3d

Tides = the daily rise and fall of the ocean, caused mostly by the **Moon's gravity**. The Moon pulls ocean water into a **bulge** on the side facing it (plus one on the opposite side). Under a bulge = **high tide**; between bulges = **low tide**. As Earth rotates through the bulges, most beaches get **2 high tides and 2 low tides each day**.

The **Sun's gravity helps too**, but it's weaker because the Sun is so far away. When the Sun and Moon line up, their pulls combine → extra-large tides!

ACTIVITY 4: Wave, Current, or Tide?

Write **W** (wave), **C** (current), or **T** (tide):

- Caused mostly by the Moon's gravity _____
- Moves energy while the water stays in place _____
- A steady "river" of water flowing in the sea _____
- The beach water creeps up, then pulls back daily _____
- Moves heat around the planet, affecting climate _____
- Made by wind; great for surfing _____

I Can... (Self-Check)

- ☐ State how much of Earth is covered by water (71%)
- ☐ Break down salt (97%) vs. fresh (3%) water
- ☐ Explain where most fresh water is stored (ice!)
- ☐ Name and order the steps of the water cycle
- ☐ Explain that the SUN powers the water cycle
- ☐ Identify the 5 ocean floor features
- ☐ Tell waves (energy) from currents (water moving)
- ☐ Explain what causes the tides (Moon's gravity)